PROJECT REPORT ON

The Driving Factors for Integrating AI in the Sustainable Development of Indian SMEs

By

Sontu Ball

Masters of Business Administration - IT

Department of Business Administration

University Of Kalyani

Kalyani 741235, India

&

Bachelor of Technology - ECE

Swami Vivekananda Institute of Science & Technology

Dakshin Gobindopur, Rajpur Sonarpur, West Bengal 700145

Abstract

Small and Medium Enterprises (SMEs) play a crucial role in the financial development and job creation within a developing country like India. Now days maintaining environmental sustainability is a big challenge like resource utilization, emissions control and increasing production efficiency for business's specially for SMEs to grow economically. By leveraging SME data along with AI technology, these businesses have the potential to gain valuable insights into their customers and competitors who can help achieving sustainable development. AI has the ability to learn from data automatically, providing SME managers with knowledge that enables better decision-making and process automation. In this research, the authors utilized Artificial Intelligence (AI) and Machine Learning (ML) technique along with other ML techniques to assess sustainability goals within Indian SMEs by the help of operational data to generate valuable insights. By combining SME data with statistics (Exploratory Data Analysis (EDA), correlation analysis), AI models, machine learning algorithms (Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and K-Means Clustering) SME managers can identify gaps in their operations like AI adoption in machines, forecasts carbon emissions, downtime and compares sustainability performance between AI-optimized and non-AI SMEs that help in making informed strategic decisions to enhance profits and achieve long-term sustainability objectives. The integration of historical SME data (secondary data) with AI technology can contribute to creating a more sustainable future. The overall research demonstrates how effectively using AI technology can enhance the sustainability of Indian SMEs.

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Chapter 1: Introduction

1.1 Introduction

In this era of digital transformation, Artificial Intelligence (AI) has emerged as a game-changer, driving operational efficiency, cost reduction, and strategic decision-making across every industry. In business, it is also effectively used daily to solve various tasks, thereby saving both time and cost and providing effective results. Among these, Small and Medium Enterprises (SMEs) play a particularly vital role in developing economies like India, contributing significantly to GDP growth, employment generation, and overall development of the country. In India, as most businesses are SME-based, they are considered the economic backbone of the country. Because SMEs have limited resources and capital, lack of skilled human resources, and resistance to change they've found it difficult to adopt AI technology in their businesses. Operating AI requires highly skilled professionals, which is often too costly for SMEs. This research investigates the impact of AI integration in SMEs by analyzing operational data from AI-enabled and non-AI-enabled machines. The focus is on understanding how AI affects key sustainability indicators such as energy consumption, material usage, waste generated, carbon emissions, and production capacity. These elements are crucial for gaining deeper insights into machine utilization, reducing production costs, and ultimately fostering business growth. The project leverages machine learning models and statistical analysis techniques to explore whether AI optimization contributes to sustainable production without compromising output performance. With increasing global pressure on industries to adopt greener and smarter manufacturing practices, this study provides timely insights into how AI can be a strategic tool for sustainable growth in the SME sector. To address this, the study applies a combination of Exploratory Data Analysis (EDA), statistical testing (e.g., t-tests), and six machine learning models—Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and K-Means Clustering. The dataset used for this analysis is sourced from Kaggle (https://www.kaggle.com/), comprising secondary data related to machine performance and operational metrics of SMEs. The models are evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix, while clustering helps identify distinct sustainability behavior patterns among AI and non-AI machines. The

project is being developed using Python, an open-source platform, within the Jupyter Anaconda environment. Jupyter Notebook, an open-source web application, facilitates the creation and sharing of documents that integrate live code, visualizations, text, and equations. Anaconda serves as a graphical user interface (GUI) navigator tool, simplifying the installation and configuration of environments. It allows for the execution of code within isolated Conda-Python environments directly from a notebook.

For data processing and model predictions, the author imported essential Python packages, including NumPy, Pandas, Matplotlib.pyplot, Seaborn, scikit-learn, and SciPy. A brief discussion of these packages is provided in Table 1.

Table 1: ML Libraries and Their Roles in the Project:

Library	Primary Use	Built On	Application Areas
NumPy	Numerical computing, multi-dimensional arrays	Python	Mathematics, Data Manipulation
Pandas	Data manipulation, cleaning, analysis	NumPy	Data Science, Analytics
Matplotlib.pyplot	2D plotting, visualizations	NumPy, Matplotlib Core	Data Visualization
Seaborn	Statistical visualizations	Matplotlib	Exploratory Data Analysis, Statistics
SciPy	Scientific computing, optimization, algebra	NumPy	Engineering, Simulation, Math Analysis
Scikit-learn	Machine learning models and algorithm	NumPy, SciPy	Classification, Clustering, Regression

Authors include waste generation, carbon emissions, material usage, production capacity, and energy consumption as key **sustainability indicators**. These indicators offer crucial insights into both environmental performance and operational efficiency.

The analysis reveals that AI-enabled machinery significantly reduces carbon emissions and waste generation compared to non-AI counterparts. While there is a slight increase in energy consumption, this occurs with no notable loss in production capacity. This project aims to bridge the gap between AI potential and its practical application in SMEs by providing evidence-based insights into how AI adoption can enhance sustainability. The outcomes are expected to assist SME managers, policymakers, and industry stakeholders in making informed decisions about integrating AI technologies for long-term, cost-effective, and sustainable development.

1.2 Objective of the project

i. To assess AI-enabled machines that can reduce carbon emissions, waste generation and energy consumption without affecting production output in SMEs.

ii. To identify key operational and environmental indicators influencing AI adoption in SMEs.

Chapter 2: Literature Review

2.1 Literature Review

Literature Review

K. Maran et al. (2023): Big Data Analytics (BDA) plays a significant role in businesses to generate, manage, and interpret huge numbers of different types of data like structured, semi-structured, and unstructured data to get various dimensional views of the business and that is very helpful in decision making. Big data mainly describes using 3 Vs – Volume, Velocity, and Variety. BDA includes artificial intelligence (AI) and machine learning (ML) to help in real-time risk assessment and increase supply chain transparency to improve scalability and accuracy. Big data analytics have evidence-based decision-making vital in complex issues, like resource optimization, waste reduction, etc.

Dr. K. Bhavana Raj et al. (2023): A combination of artificial intelligence (AI), machine learning (ML), and the Internet of Things (IoT) in Industry 4.0 is an opportunity for SMEs to grow their business that has already been adopted by large firms and achieved fruitful results. Most of the SMEs are not properly adopted because they have infrastructure gaps, skill shortages, and financial constraints. In this paper, authors integrate an AI-based model for SMEs that can help reduce the uncertainty in sales forecasting and also enhance decision-making, but most SMEs don't know how to use this system to improve performance. The authors also discuss the importance of a structured performance framework integrating AI-driven modules for complex decision-making and personalized offerings that can be easily adaptable if SMEs provide simplified access, training, and target usage.

Silva, Joaquim P. et al. (2022): The Integration of Artificial Intelligence (AI) and management analytics can provide a better result for any type of organization but due to lack of technical expertise, high cost, low capital, and poor accessibility, SMEs have only limited access which is totally opposite for large enterprises, so they can easily adopt this, maintain this and enhance their decision making. According to the authors, process standardization will help SMEs with AI and analytics adoption because business analytics and decision support systems (DSS) need a structured environment. As AI costs decrease,

SMEs should be more open to adoption but to achieve success they have to use high-quality data and standardized frameworks.

Chiang Liang Kok et al. (2024): Artificial intelligence (AI) and Machine learning (ML) play a very important role for SMEs but they need specialized technical skills and huge financial investment that barriers can be reduced by these three prominent low-code or no-code AI platforms like Microsoft Power Platform(Power Apps, Power Automate, and Power BI), Google AutoML(machine learning model: image classification and natural language processing), and Amazon SageMaker(model deployment, data science expertise). The enterprise can select this suitable tool depending on their specific needs, technical proficiency, and budget. SMEs can now easily create powerful AI models with minimal technical expertise by using platforms. They also play a significant role in bridging the digital divide for SMEs.

Giridhar Mallela et al. (2024): For a developing country's economic development, SMEs play a vital role but most of the time they face barriers to internationalization due to limited resources, high implementation costs, technical complexity, and data privacy. Edge AI technologies like smart sensors, autonomous systems, and industrial IoT can help in overcoming challenges, real-time decision-making, improving operational inefficiencies like supply chain management, meeting local regulatory demands, etc. The authors used two datasets “AI_Tech_Adoption_SME” and “AI_Tech_Benefits_SME” that collect from sectors like manufacturing, finance, and healthcare. Edge AI integration in SMEs helps to develop cost-effective, scalable solutions and supportive policy frameworks.

Biswajit Biswas et al. (2025): Sales forecasting is very crucial for every business, especially for e-commerce where we work with large and complex datasets, where we cannot use statistical techniques but easily use machine learning (ML) techniques to identify patterns and relationships like consumer behaviour and enhance sales strategies. ML models help decode customer purchase intentions and implement marketing strategies more effectively. The authors discuss that they used linear regression, Decision trees, and Random Forest models but Random Forest models offered the best performance in sales forecasting tasks, especially in retail and supermarket environments. This paper shows machine learning in e-commerce helps in a real-world data-driven approach to sales prediction of mobile phones.

Robert g. Cooper (2024): The use of Artificial Intelligence (AI) is rapidly growing day by day. Larger companies are already implementing AI in many sectors like product development, such as market analysis, design, testing, and project management, and getting better results and performance, especially in new product development (NPD). Based on the Authors SMEs only adopt 15% of AI in their NPD than larger firms, and this adaptation is very low in early stages like idea generation and market research where they need this mostly. Factors like trust in AI, senior management support, clear performance benefits, low awareness, and readiness are needed in AI adoption, for these SMEs face slow progress. This study tells us more education, leadership commitment will help SMEs adapt to AI tools.

T A Han et al. (2023): AI tools sometimes show biased outcomes or act unpredictably, if they are not designed with proper ethical and safety measures. Maintaining secure environments, transparency, and fairness authors create a risk evaluation tool using explainable AI (XAI) for SMEs. By using it, SMEs can assess these four major areas of risk—software, hardware, ethical, and legal in different sectors like healthcare, agriculture, education, and business. This research talks about AI governance and AI adaptation by offering a practical and explainable framework that is really needed by SMEs even without deep technical knowledge but initially needs training and expert guidance.

Giridhar Mallela et al. (2024): Artificial Intelligence (AI) helps small and medium enterprises (SMEs) in studying the global market by analyzing large sets of data to uncover patterns and predictions. AI tools help organizations identify customer preferences and market trends so they can provide personalized services, improve customer satisfaction, choose the best locations for expansion, optimize delivery routes, reduce costs, etc. The authors also describe the securities and skilled professionals that are very important in implementing AI tools. This paper mainly discusses AI-powered geospatial analysis that can be helpful for SMEs to find high-potential markets and reduce risks during international expansion.

Lynn Miller et al. (2020): Artificial Intelligence (AI) can be a powerful tool to analyze progress on the United Nations Sustainable Development Goals (SDGs). In traditional methods data collection is incomplete, inconsistent, and unreliable, whereas data collection with the help of AI and ML is faster, more accurate, and scalable, it is used to measure Sustainable Development Indicators (SDIs). Past time SDG monitoring time used satellite

imagery and remote sensing, especially in sectors like and use, water quality, and poverty detection are too challenging but now using “Explainable AI (XAI)” we are getting more understandable and actionable results in policy making. The authors said that integrating AI with policy development requires technical systems and social governance structures but it is very effective in real-world sustainable development policy generation.

2.2 Literature Gap

From all this literature review, author found that most of the authors just concentrated on exploratory research methods or review based research, low code platforms that is highly costly and they didn’t create any model using code and open-source platform also because SMEs has limited resources and expert manpower. The most of the authors used data of big organization and mostly secondary data because the SMEs data is not that much available and it’s also difficult to collect data from SMEs. So not much study has been done on SMEs to implement AI and ML related analysis for sustainable development.

Chapter 3: Research Methodology

3.1. Research Design

This research follows a quantitative approach using exploratory data analysis and machine learning modelling. The goal is to analyse operational data from Indian SMEs to understand how the integration of Artificial Intelligence (AI) in machinery affects carbon emissions, waste generation, and overall sustainability.

3.2. Data Collection

- **Type of Data**: Secondary data
- **Source**: Authors collect the dataset from open-source platform Kaggle (a platform where different types of data are available for data scientists and machine learning practitioners). Dataset that includes details such as:
  i. Factory ID
  ii. Machine ID
  iii. Energy consumption
  iv. Material usage
  v. Waste generated
  vi. Carbon emissions
  vii. Production capacity
  viii. Whether AI optimization was applied

- **Dataset Size**: Authors took 228 data from this dataset, where 114 data where AI_Optimization_Applied = 'Yes' and others 114 'AI_Optimization_Applied' = No.

Table 2: Dataset Sample:

Factory ID	Machine ID	Energy Consumption	Waste Generated	Carbon Emission	Production Capacity	Material Usage	AI Optimization Applied
F19	M239	166	10	71	286	449	Yes
F8	M157	242	26	59	259	324	No
F4	M319	170	24	79	252	295	No
F7	M338	248	11	30	293	409	Yes
F4	M426	207	11	61	159	405	Yes

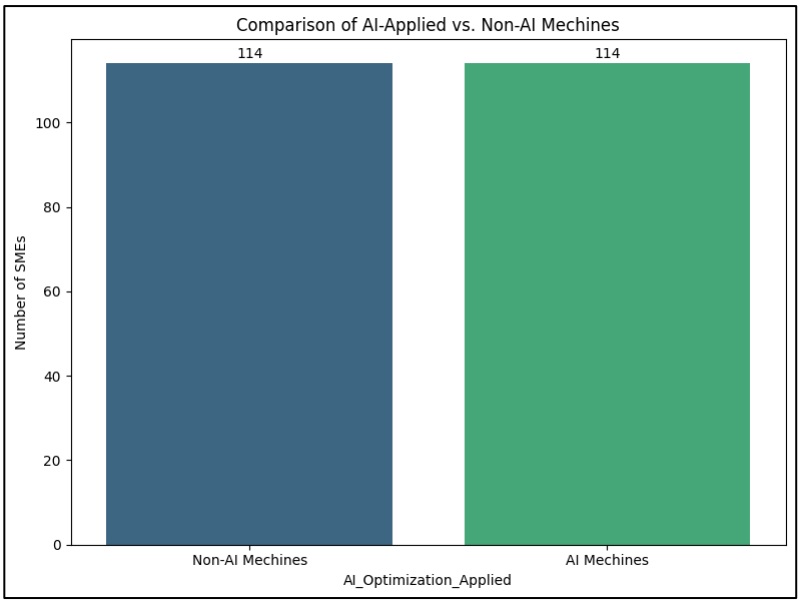

Figure 3.1: Bar Plot: SMEs with and without AI.

3.3. Data Preprocessing

Categorical variables like AI_Optimization_Applied were encoded (Yes = 1, No = 0).

Table 3: Categorical to Numerical Conversion:

Factor Y ID	Machine ID	Energy Consumption	Waste Generated	Carbon Emissions	Production Capacity	Material Usage	AI Optimization Applied
F19	**M239**	166	10	71	286	449	1
F8	**M157**	242	26	59	259	324	0
F4	**M319**	170	24	79	252	295	0
F7	**M338**	248	11	30	293	409	1
F4	**M426**	207	11	61	159	405	1

- Missing and inconsistent values were cleaned manually or imputed.

Table 4: Missing Values in the Dataset:

Name	Null Value
Factory ID	0
Machine ID	0
Energy Consumption	0
Waste Generated	0
Carbon Emissions	0
Production Capacity	0
Material Usage	0
AI Optimization Applied	0

3.4. Feature Selection

Selected features based on relevance to **sustainability indicators**:

i. Energy_Consumption
ii. Material_Usage
iii. Waste_Generated
iv. Carbon_Emissions
v. Production_Capacity

These features serve as independent variables (X), and the target variable (y) is whether AI Optimization was applied.

3.5. Train-Test Split

- The dataset was divided using train_test_split () from sklearn:
    - 80% used for training
    - 20% used for testing
- Random state was set to 42 for reproducibility.

3.6. Modelling Techniques

To assess how well AI adoption could be predicted and to explore its relation to sustainability metrics, the following machine learning models were applied:

i. Logistic Regression
ii. Support Vector Machine (SVM)
iii. K-Nearest Neighbors (KNN)
iv. Decision Tree
v. Random Forest
vi. K-Means Clustering (for unsupervised grouping)

Each model was evaluated based on accuracy, confusion matrix, precision, recall, and graphical visualization.

3.7. Model Evaluation Metrics

Used to assess prediction performance:

i. Classification Report: Precision, Recall, F1-Score
ii. Confusion Matrix
iii. ROC Curve and AUC Score
iv. Feature Importance (in Random Forest)
v. Cluster Quality (in K-Means)

3.8. Tools & Libraries

- Programming Language: Python
- Libraries: pandas, NumPy, matplotlib, seaborn, scikit-learn, SciPy

3.9. Ethical Considerations

Only secondary data was used, ensuring there was no personal or confidential data involved.

Chapter 4: Data Analysis and Interpretation

Python Libraries used in this project:

This section lists the essential Python libraries utilized in this project. These include packages for data manipulation, statistical analysis, data visualization, and machine learning.

- ✓ import pandas as pd
- ✓ import numpy as np
- ✓ import seaborn as sns
- ✓ import matplotlib.pyplot as plt
- ✓ from sklearn.model_selection import train_test_split
- ✓ from sklearn.preprocessing import StandardScaler
- ✓ from sklearn.linear_model import LogisticRegression
- ✓ from sklearn.tree import DecisionTreeClassifier
- ✓ from sklearn.ensemble import RandomForestClassifier
- ✓ from sklearn.svm import SVC
- ✓ from sklearn.neighbors import KNeighborsClassifier
- ✓ from sklearn.cluster import KMeans
- ✓ from sklearn.metrics import classification_report, confusion_matrix
- ✓ from scipy.stats import mode
- ✓ from scipy.stats import ttest_ind

Figure 4.1 presents a bar plot illustrating the number of factories and the corresponding number of machines available in each factory.

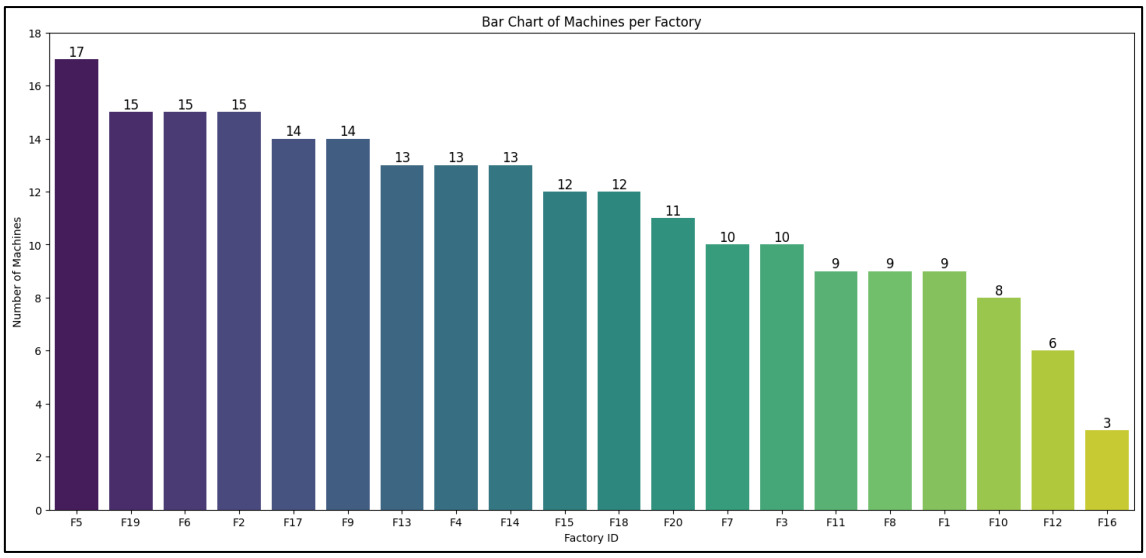

Figure 4.1: Bar Chart of Machines per Factory.

Table 5 presents the statistical description of the dataset, providing key values such as count, mean, standard deviation, minimum, and maximum.

Table 5: Descriptive Statistics of the Dataset

	Energy Consumption	Waste Generated	Carbon Emissions	Production Capacity	Material Usage
count	228.000000	228.000000	228.000000	228.000000	228.000000
mean	206.000000	28.942982	54.942982	222.464912	357.052632
std	54.631485	10.238620	13.657590	42.835446	88.088196
min	102.000000	10.000000	30.000000	150.000000	200.000000
25%	160.000000	21.000000	44.750000	180.500000	281.000000
50%	215.000000	29.000000	54.500000	228.500000	360.500000
75%	250.250000	37.000000	67.000000	257.250000	436.250000
max	299.000000	49.000000	79.000000	298.000000	497.000000

Figures 4.2, 4.3 and 4.4 present a Box plot, Histogramand Correlation matrix, respectively. These visualizations depict the relationships among Energy Consumption, Waste Generated, Carbon Emissions, AI Optimization Applied, and Production Capacity. Such graphical representations offer concise insights into the dataset's characteristics.

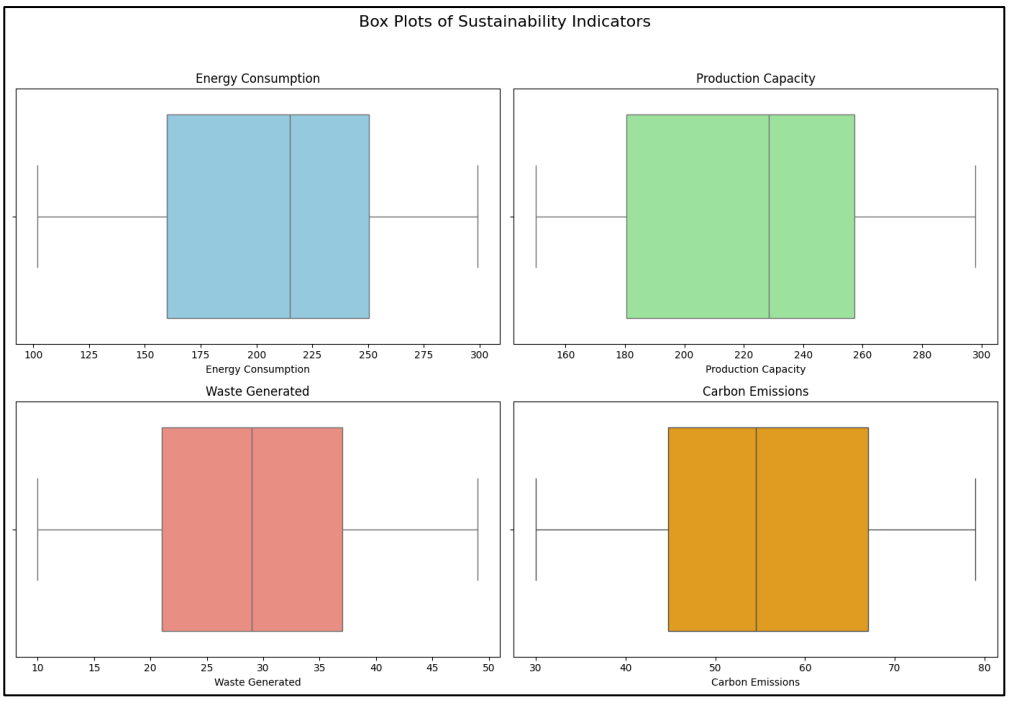

Figure 4.2: Box plot of Sustainability Indicators.

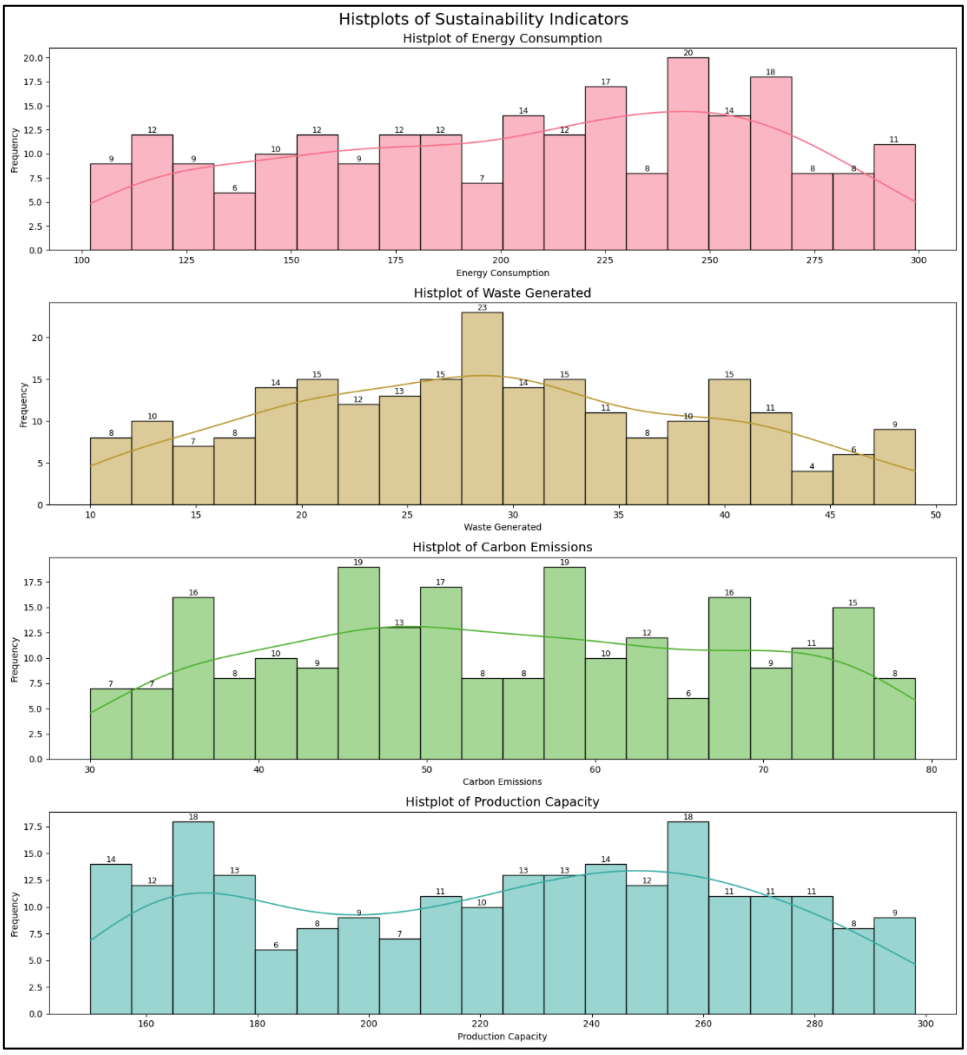

Figure 4.3: Hist plot of Sustainability Indicators.

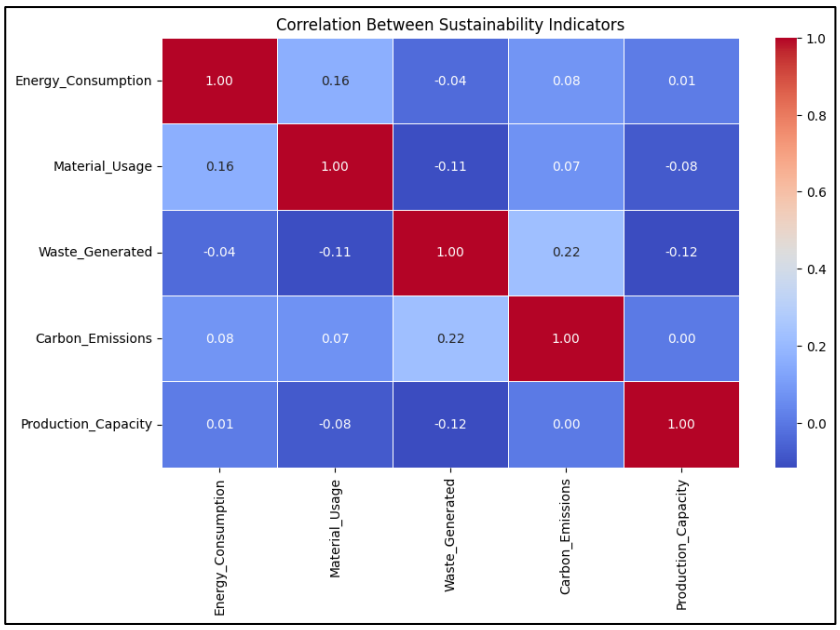

Figure 4.4: Correlation matrix of Sustainability Indicators.

This section presents key insights derived from statistical analysis, model performance, and clustering, emphasizing the sustainability differences between AI-optimized and non-AI machines in Indian SMEs. The average comparison (Table 6) reveals that AI integration reduces carbon emissions and waste without compromising production capacity. T-tests and hypothesis testing (Table 7) confirm that these reductions are statistically significant, while energy usage and production output remain unaffected. Visual analyses, including box plots and histograms (Figure 4.6), highlight distinct differences in the distribution of emissions and waste, and the pair plot (Figure 4.8) shows AI machines forming more efficient clusters. The bar plot (Figure 4.5) further reinforces that AI machines consistently outperform non-AI counterparts on sustainability metrics. Clustering analysis (Table 9 and Figure 4.8) using K-Means groups machines into three clusters based on sustainability, with AI-optimized machines clearly forming low-emission, low-waste clusters. These findings collectively demonstrate that AI adoption leads to measurable environmental benefits—reduced emissions and waste—without affecting productivity, with statistical validation and clustering confirming the consistent, efficient behavior of AI-enhanced machines.

Table 6: Average Values of Sustainability Indicators: AI vs. Non-AI Machines

SN	*Parameters*	*AI machines*	*Non-AI machines*	*Interpretation*
01	**Carbon Emissions**	48	60.50	AI machines emit significantly less carbon, indicating improved environmental efficiency.
02	**Production Capacity**	228.5	228	Almost the same, suggesting AI improves sustainability without compromising output.
03	**Energy Consumption**	215	211	Slightly higher in AI machines, possibly due to additional computational tasks.
04	**Waste Generated**	23	33	AI machines produce less waste, improving resource efficiency.

Table 7: T-Test and Hypothesis Results of Sustainability Indicators: AI vs. Non-AI Machines

SN	PARAMETERS	T – TEST	P-VALUE	SIGNIFICANCE
01	**Carbon Emissions**	-5.3457	0.000000220081092	Significant difference — AI machines have lower emissions.
02	**Production Capacity**	-0.0370	0.970499003654910	Highly significant — AI machines generate much less waste.
03	**Energy Consumption**	1.0452	0.297063981526186	Not statistically significant.
04	**Waste Generated**	-8.2537	0.000000000000013	No significant difference.

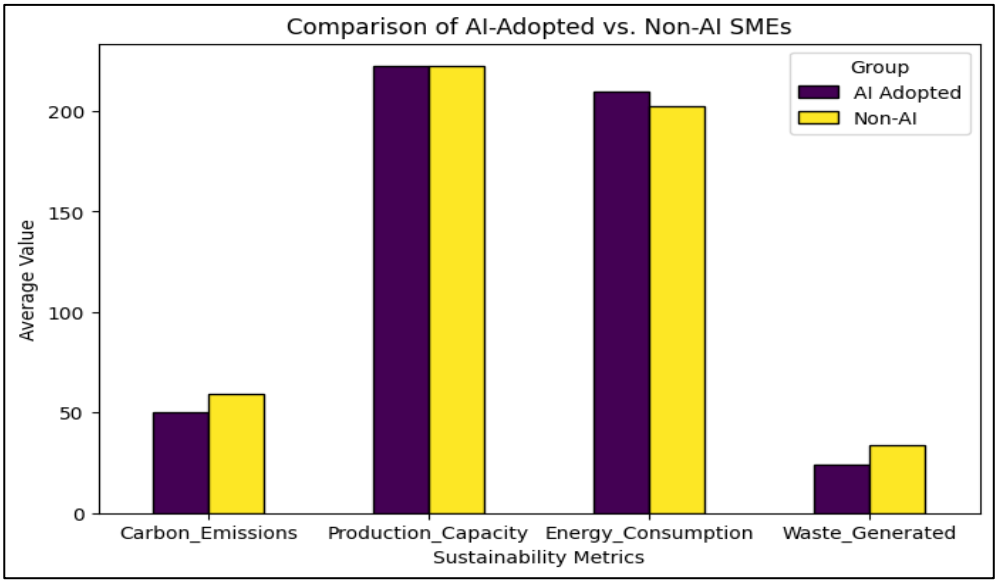

Figure 4.5: Bar Chart of Sustainability Indicators: AI vs. Non-AI Machines.

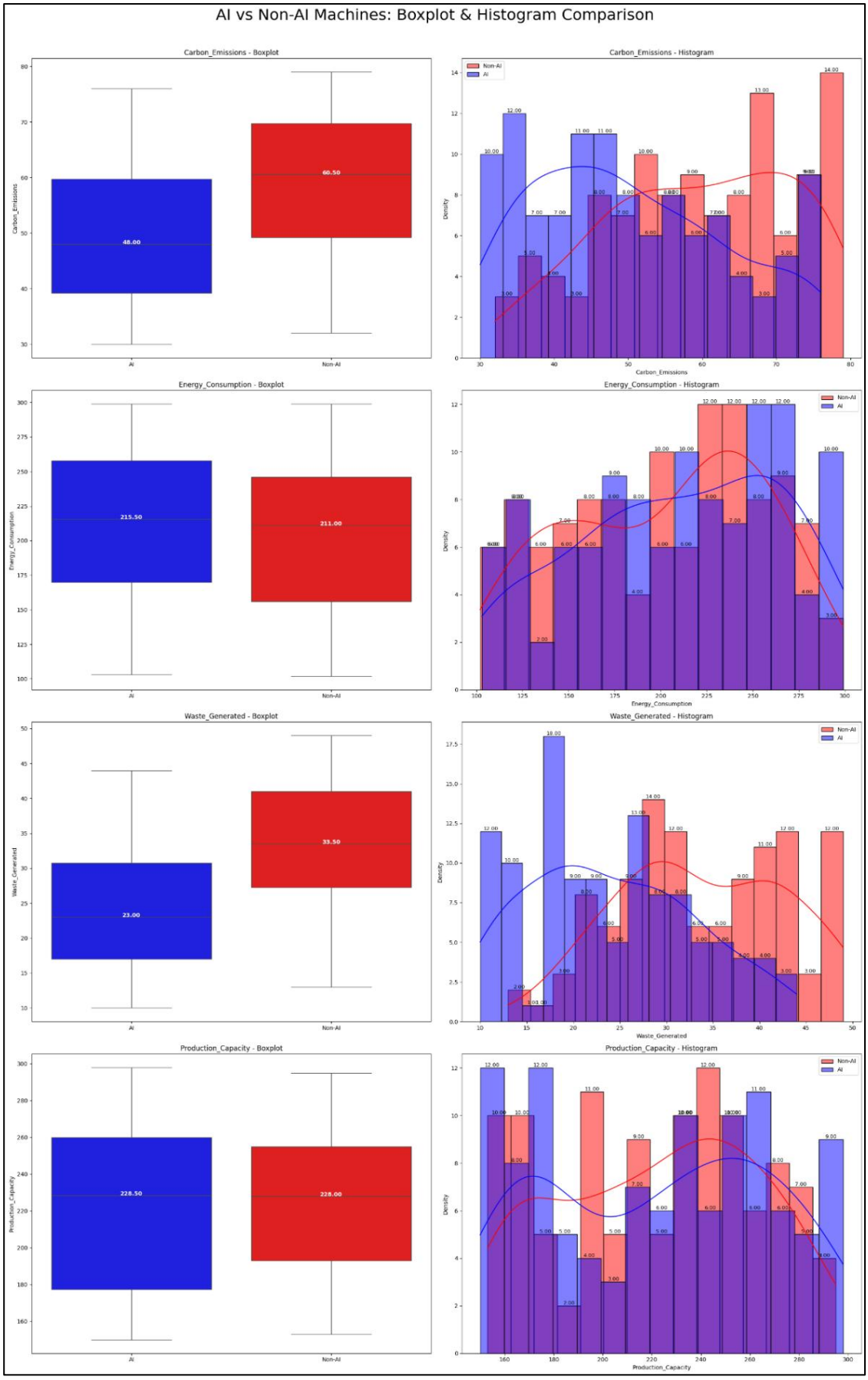

Figure 4.6: Box and Histogram Plots of Sustainability Indicators: AI vs. Non-AI Machines

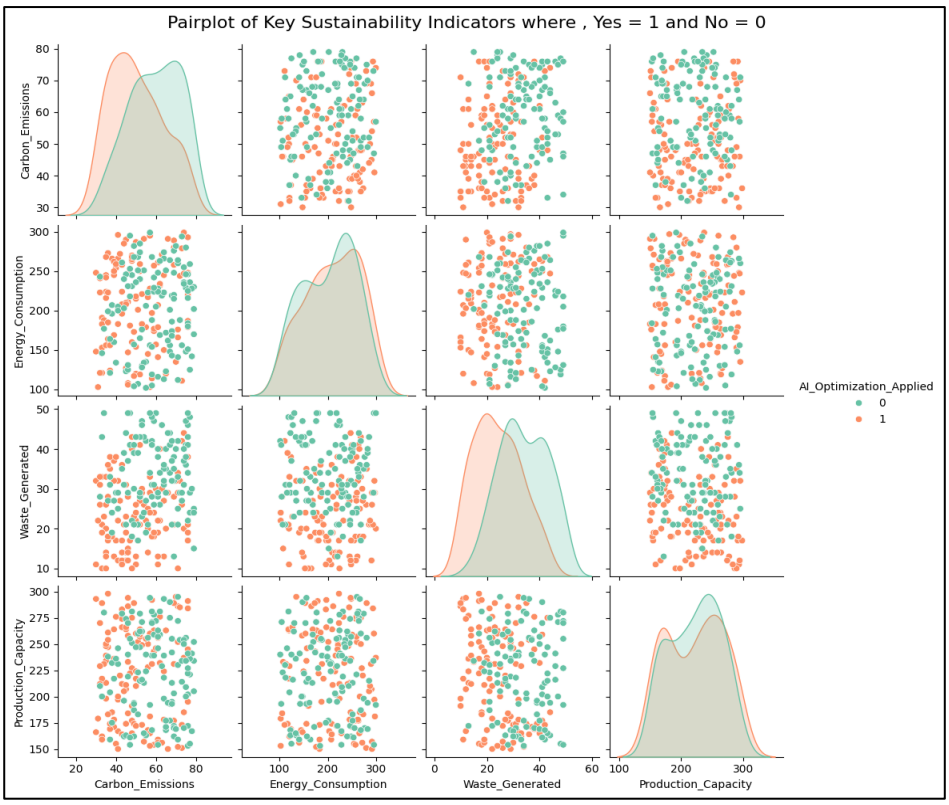

Figure 4.7: Pair plot of Sustainability Indicators: AI vs. Non-AI Machines

Table8: Cluster Analysis of sustainability indicators:

Cluster	*Energy Consumption*	*Waste Generated*	*Carbon Emissions*	*Interpretation*
0	155.016667	36.033333	57.850000	non-AI, high waste
1	199.855556	21.677778	46.155556	AI-optimized, efficient
2	252.307692	31.871795	62.846154	high-performance

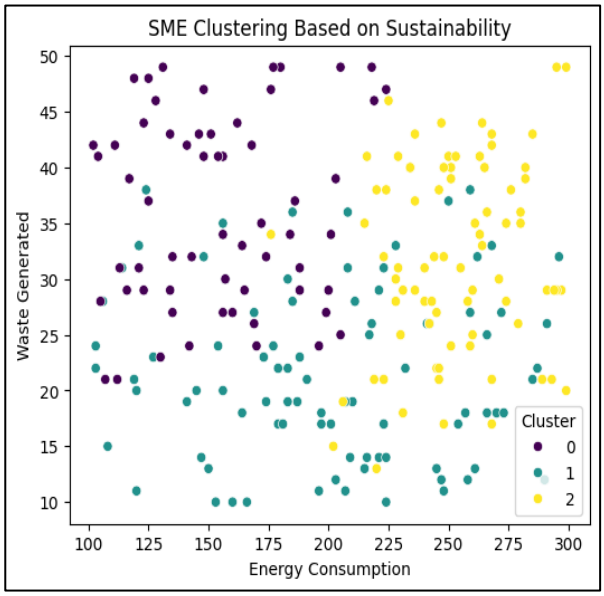

Figure 4.8: Cluster Analysis: Waste Generated vs. Energy Consumption

This section compares the performance of six machine learning models used to predict AI adoption in Indian SMEs based on sustainability indicators. Logistic Regression performed best, with 83% accuracy, high precision (0.91), and strong recall (0.90), making it highly reliable for decision-making. Decision Tree (80%) and Random Forest (78%) also showed solid performance. However, SVM (72%) and KNN (70%) were less effective, while K-Means Clustering, as an unsupervised method, had the lowest accuracy (59%) and was unsuitable for precise classification.

Table09: Comparison of Machine Learning methods and performance metrics:

Sn	ML Methods	Precision	Recall	F1-Score	Support	Class	Accuracy
01	Logistic Regression	0.91	0.77	0.83	26	0	0.83
		0.75	0.90	0.82	20	1	
02	SVM Classifier	0.78	0.69	0.73	26	0	0.72
		0.65	0.75	0.70	20	1	
03	KNN Classifier	0.77	0.65	0.71	26	0	0.70
		0.62	0.75	0.68	20	1	
04	Decision Tree	0.90	0.73	0.81	26	0	0.80
		0.72	0.90	0.80	20	1	
05	Random Forest	0.83	0.77	0.80	26	0	0.78
		0.73	0.80	0.76	20	1	
06	K-Means Clustering	0.67	0.54	0.60	26	0	0.59
		0.52	0.65	0.58	20	1	

Table 10: Comparison of Machine Learning Models by Accuracy:

Model	Accuracy	Comments
Logistic Regression	**83%**	• Best performing overall. • Suitable for in this case.
SVM	**72%**	• Lower accuracy, precision & recall. • Not suitable for this type of problem.
KNN	**70%**	• Lowest accuracy. • Not suitable for this type of problem.
Decision Tree	**80%**	• Balanced performance. • Slightly more mistakes than LR. • Good if you want decent performance.
Random Forest	**78%**	• Fair balance accuracy. • Reliable, but less accurate than Decision Tree
K-Means Clustering	**59%**	• Lower performance overall. • Not preferred in this case.

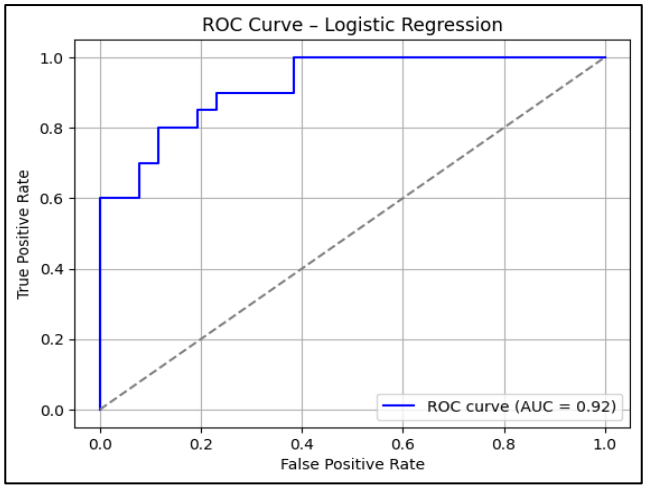

Figure 4.9: Logistic Regression Graph

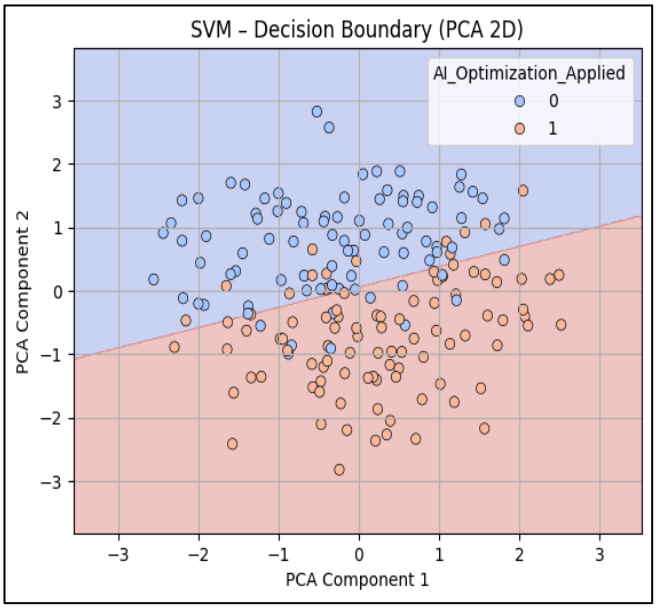

Figure 4.10: SVM Classifier graphs.

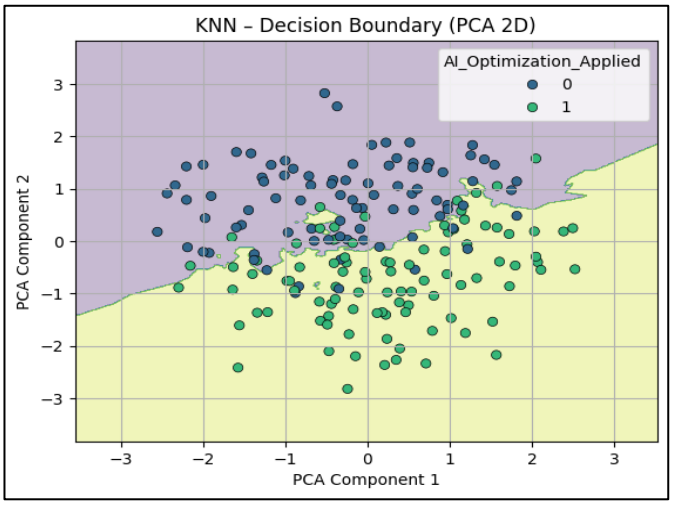

Figure 4.11: KNN Classifier Graphs.

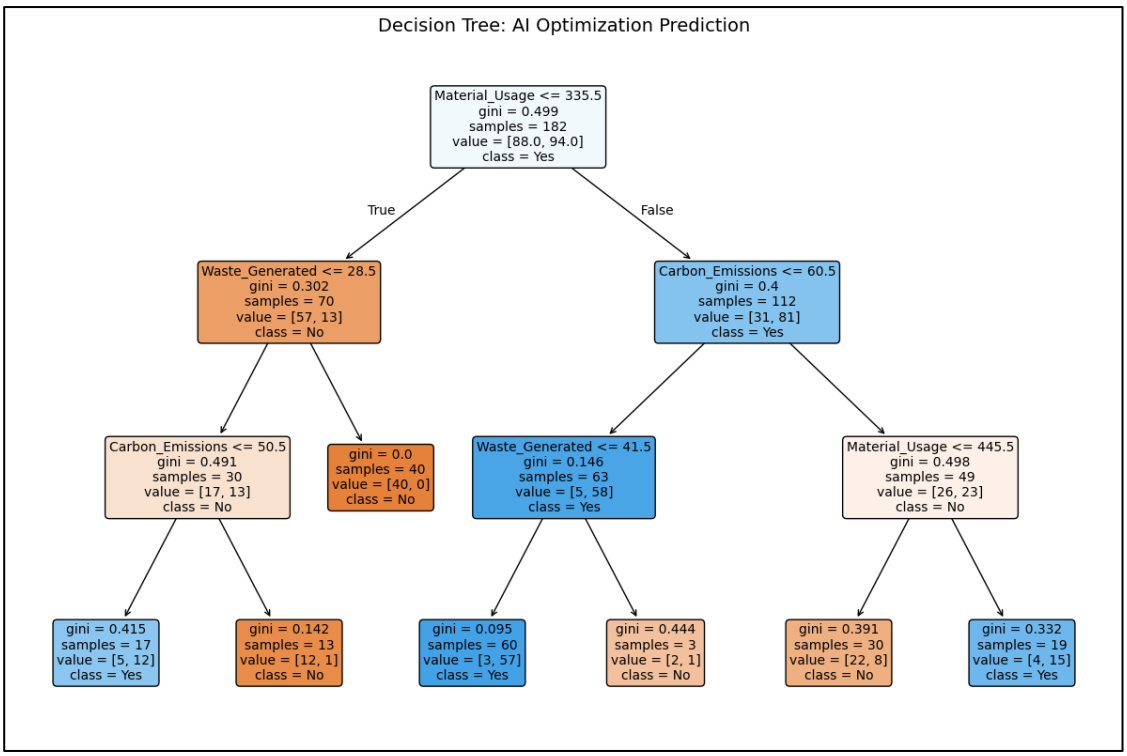

Figure 4.12: Decision Tree Graphs

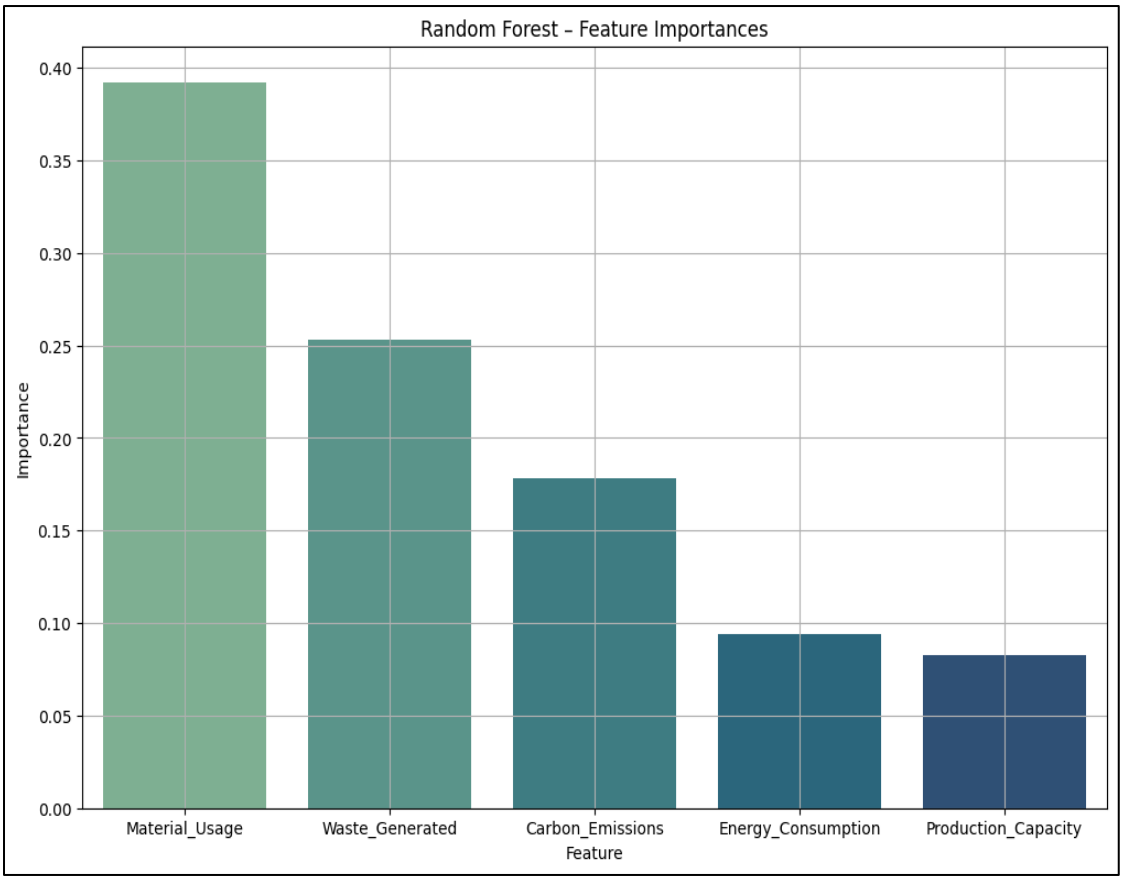

Figure 4.13: Random Forest Graphs.

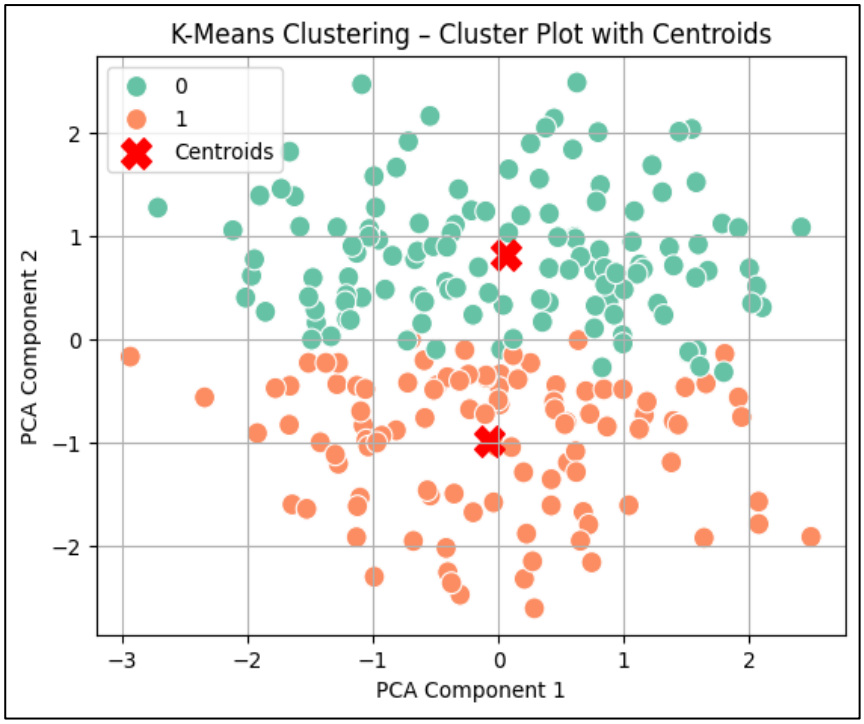

Figure 4.14: K-Means Clustering graphs.

Analysing results of this six ML models author get most accuracy using logistic regression. The Logistic Regression model is the best for your SME dataset to predict AI adoption. It accurately identifies companies using or not using AI optimization with 83% accuracy, making it reliable for decision support or policy planning in sustainable development. So, for getting more insights author create confusion matrix using logistic regression.

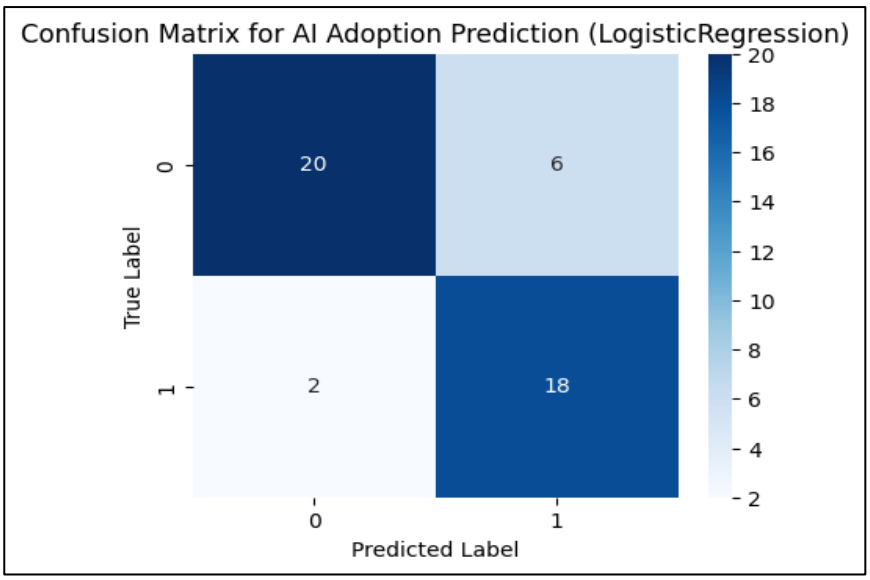

Figure 4.15: Confusion Matrix for Logistic Regression Model.

In '**Figure 4.15**' the results get from the confusion matrix logistic regression classifier reveal how well the model predicts whether SMEs have applied AI Optimization based on

sustainability indicators (like energy use, waste, carbon emissions, etc.). The results and its discussions are showing in the 'Table 11& Table 12'.

Table 11: Confusion Matrix results:

	Predicted: No AI	***Predicted: AI***
Actual: No AI	20 (True Negatives)	6 (False Positives)
Actual: AI	2 (False Negatives)	18 (True Positives)

Table 12: Discussion of Confusion Matrix Results:

Predictions	Value	Comments
True Negatives	**20**	The model reliably distinguishes non-AI users, which helps in targeting them for potential AI adoption initiatives.
False Positives	**6**	The model might have confused some efficient or sustainable SMEs as AI users even though they aren't.
False Negatives	**2**	A small number of AI adopters were missed.
True Positives	**18**	The SMEs that are genuinely AI-optimized, and the model was able to recognize them correctly.

The confusion matrix demonstrates that the model is effectively predicting AI adoption in SMEs using sustainability indicators. Although there are a few false predictions, the model's overall generalization performance is strong.

Chapter 5: Conclusion, Findings, Managerial Implications, Limitations and Future Scope

5.1 Findings

i. The finding will be showing that the AI integrated machines are generating low carbon emissions, waste but the energy consumption is increased.
ii. The results of AI enabling machines producing low carbon emissions, generating low waste and increasing the Production Capacity (Details shown in table number 7). Now it is understood that AI adaptation helps SMEs to decrease the environmental pollution and improving operation quality.

5.2 Conclusion

Indian Small and Medium Enterprises (SMEs) are increasingly adopting emerging technologies to enhance their operations. Modern organizations are widely leveraging Artificial Intelligence (AI) and Machine Learning (ML) to automate tasks, improve efficiency, and enhance accuracy. This study clearly shows that using Artificial Intelligence (AI) in the machines of Indian Small and Medium Enterprises (SMEs) can make their operations more sustainable by reducing energy use, generating less waste, decreasing carbon emissions, and using fewer raw materials — all without reducing companies production capacity. By analyzing the data using statistics, AI and Machine Learning (ML) techniques, our findings indicate that SMEs integrating AI into their machinery demonstrate superior environmental performance compared to those that do not. These AI-enabled SMEs demonstrate enhanced efficiency, resource conservation (evidenced by lower material usage and reduced waste generation), and diminished environmental pollution, particularly through decreased carbon emissions. The findings underscore AI's capacity to empower SME managers in identifying and rectifying operational inefficiencies. However, a notable problem is that AI machines use more energy than those without AI. Furthermore, six machine learning models were evaluated to predict AI adoption based on sustainability indicators. Logistic Regression outperformed other models, achieving 83% accuracy along with high precision (0.91) and recall (0.90), making it the most reliable for decision support and policy formulation. The successful implementation of AI not only leads to cost savings but also fosters more eco-friendly

production practices. The authors believe that future governmental initiatives in India, focused on making AI technology more affordable and accessible through enhanced training, will significantly boost AI adoption among SMEs. This, in turn, will accelerate the growth of Indian SMEs while simultaneously contributing to environmental protection and long-term national economic and sustainable development.

5.3 Managerial Implications

1. The integration of AI into SME operations helps managers make data-driven decisions, particularly in carbon emission reduction, and waste control.
2. Managers can utilize the insights from AI models (e.g., Decision Trees, SVM etc) to identify underperforming processes and reallocate energy or materials more efficiently without reducing production capacity.
3. The comparison between AI-enabled and non-AI machines provides a clear performance benchmark, helping SMEs identify the ROI of AI adoption.
4. SMEs adopting AI are better positioned to meet environmental regulations and sustainability standards, reducing financial risks.

5.4 Limitations

1. The study relied on secondary SME operational data, which may not capture real-time performance.
2. Authors only used basic ML model here, but advance ML or deep learning provide more better and accurate results.
3. In this study only one manufacture data is used, if the work done with different sectors data it will helpful for overall Indian SMEs.
4. The study assumes smooth integration of AI, but real-world adoption may face barriers like lack of technical expertise or high upfront costs in SMEs.
5. ML model results will predict better especially in large datasets buthere author used small datasets.

5.5 Future scope

1. Develop and test real-time AI dashboards for SMEs to continuously monitor emissions, waste, and energy usage.
2. Work with larger datasets especially primary data to achieve more robust results and better ML prediction.
3. Using deep learning and advanced ML techniques for developing intelligent systems and automating complex tasks, also achieving higher accuracy and efficiency
4. Extend the study by adding others sector of SMEs in related industries and segmenting SMEs by industry (e.g., textiles, manufacturing, food processing) to determine AI impact by sector.
5. Explore combining AI with IoT sensors to automate and optimize machine performance dynamically.
6. Conduct follow-up studies over multiple years to track the long-term sustainability outcomes of AI adoption.
7. Propose scalable AI literacy and training programs to equip SME employees and managers for smoother transitions.

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